



Sychel customer churn Application

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Introduction

Predictive classification project to identify customers at high risk of churn for a telecommunications stakeholder. Built with a recall-first objective to minimize missed churners, using interpretable baselines and iterative modeling with appropriate preprocessing for an imbalanced dataset.



Overview

Predict whether a customer will churn based on their usage patterns, service plans, and account information.

Business objective

- Predict churn using usage, plans, service interactions
- Handle class imbalance correctly
- Prioritize recall over accuracy
- Deliver interpretable insights

Business Problem

Customer churn is a major challenge for telecommunication companies as acquiring new customers is more costly than retaining existing ones. The customer retention team's goal is to proactively identify customers at high risk of churning and try to help before they leave.



Goal

1. identify high-risk
churn customers
early
2. Emphasis on
recall and
interpretability

Data Preparation

- The data had no missing or null values, it had a mix of numerical and categorical features with 3,333 customers
- The phone number column was dropped as it was non-predictive
- The area code was changed from object type to category

EDA

01

Class distribution (countplot of churn)

The churn class is the minority (~14.5%), confirming an imbalanced problem. Accuracy alone would be misleading; recall and PR-AUC are more appropriate for evaluation.

02

Usage distribution

Most customers exhibit moderate usage with a long tail of heavier users. This motivated log transforms for usage minutes to reduce skew and stabilize models.

06



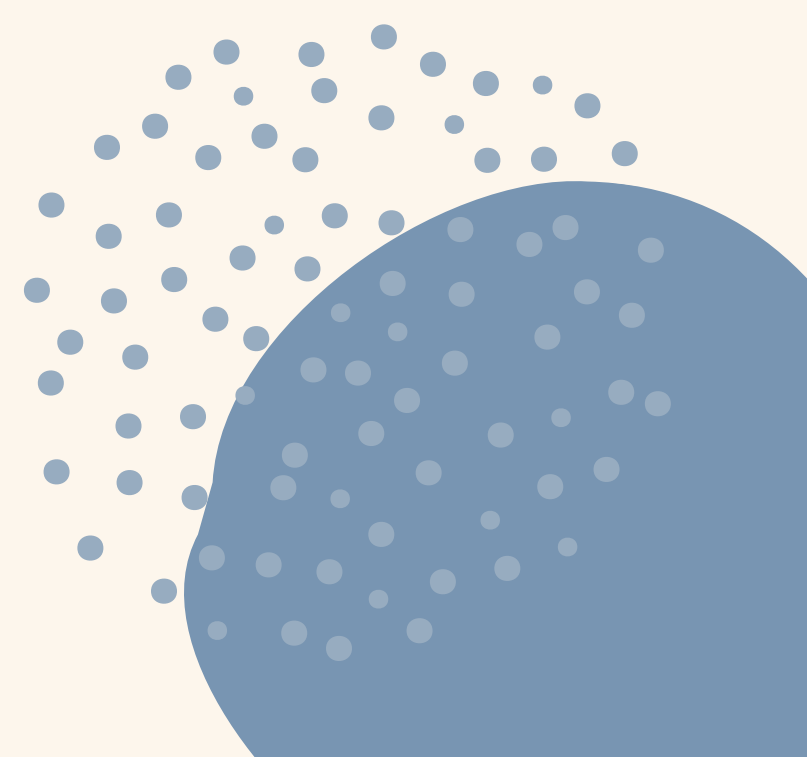
03

Customer service calls vs churn

Churners tend to have higher customer service calls, consistent with dissatisfaction. This informed creating `service_call_rate` and keeping customer service calls as key predictors.

04

Correlation heatmap

- Charges are highly correlated with minutes (by design), so charge columns were dropped to avoid overweighting the same signal.
 - Usage metrics across periods are moderately correlated; engineered averages and totals help summarize intensity cleanly.
- 

Feature engineering

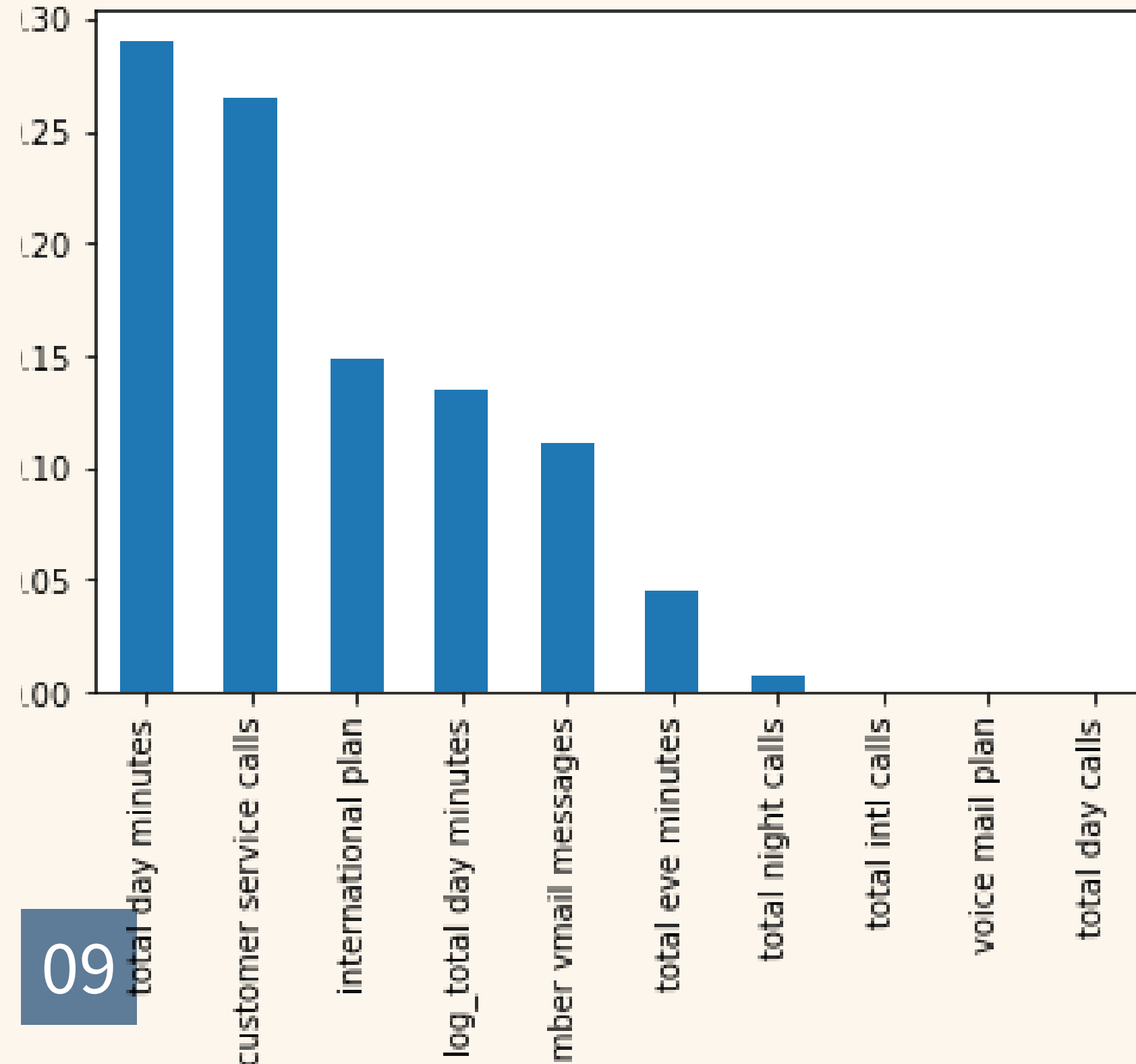
- Removed charge columns
- Created interaction features:
- Average call duration
- Total call volume
- Service call rate
- Log-transformed usage

Modeling

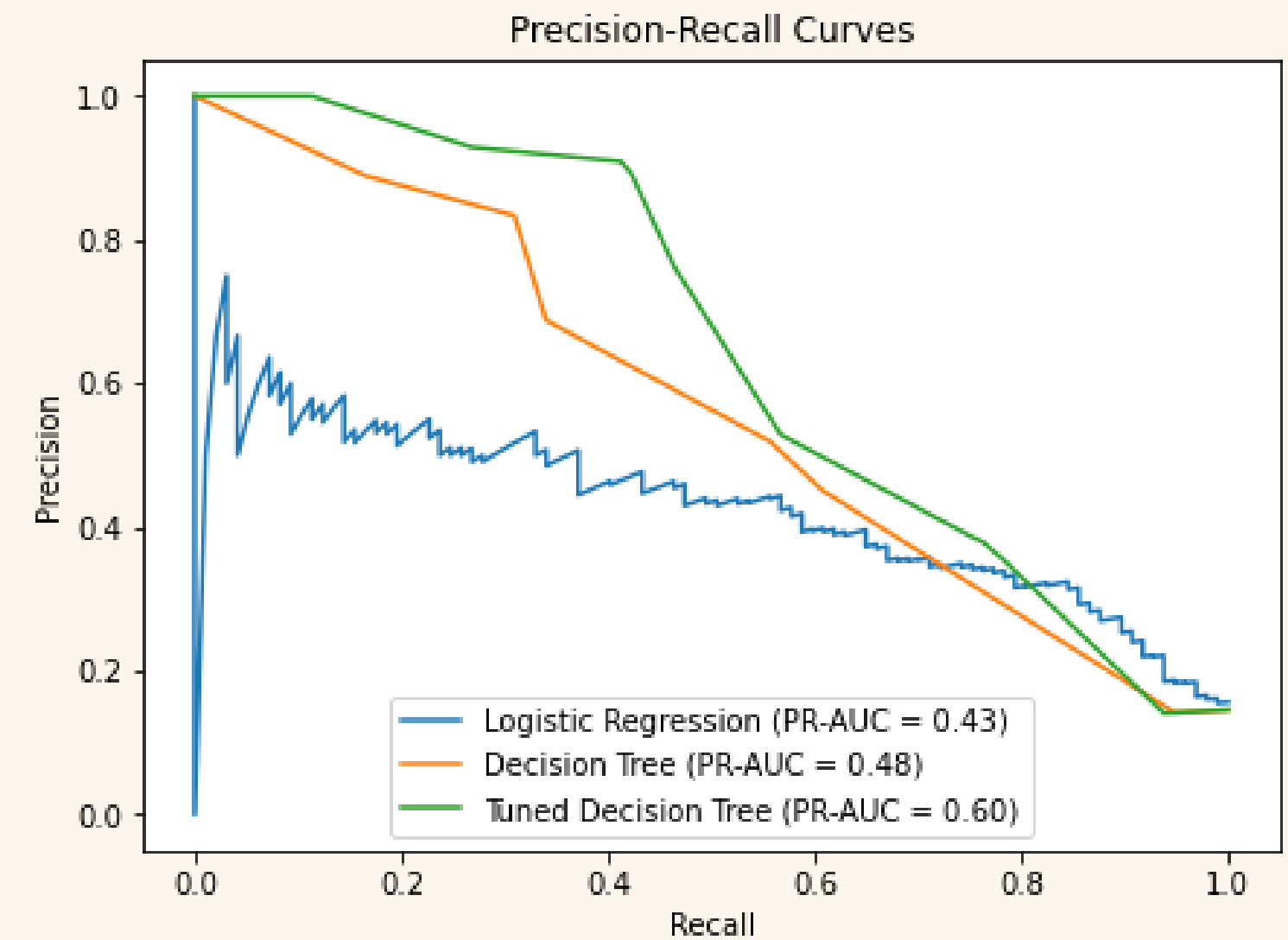
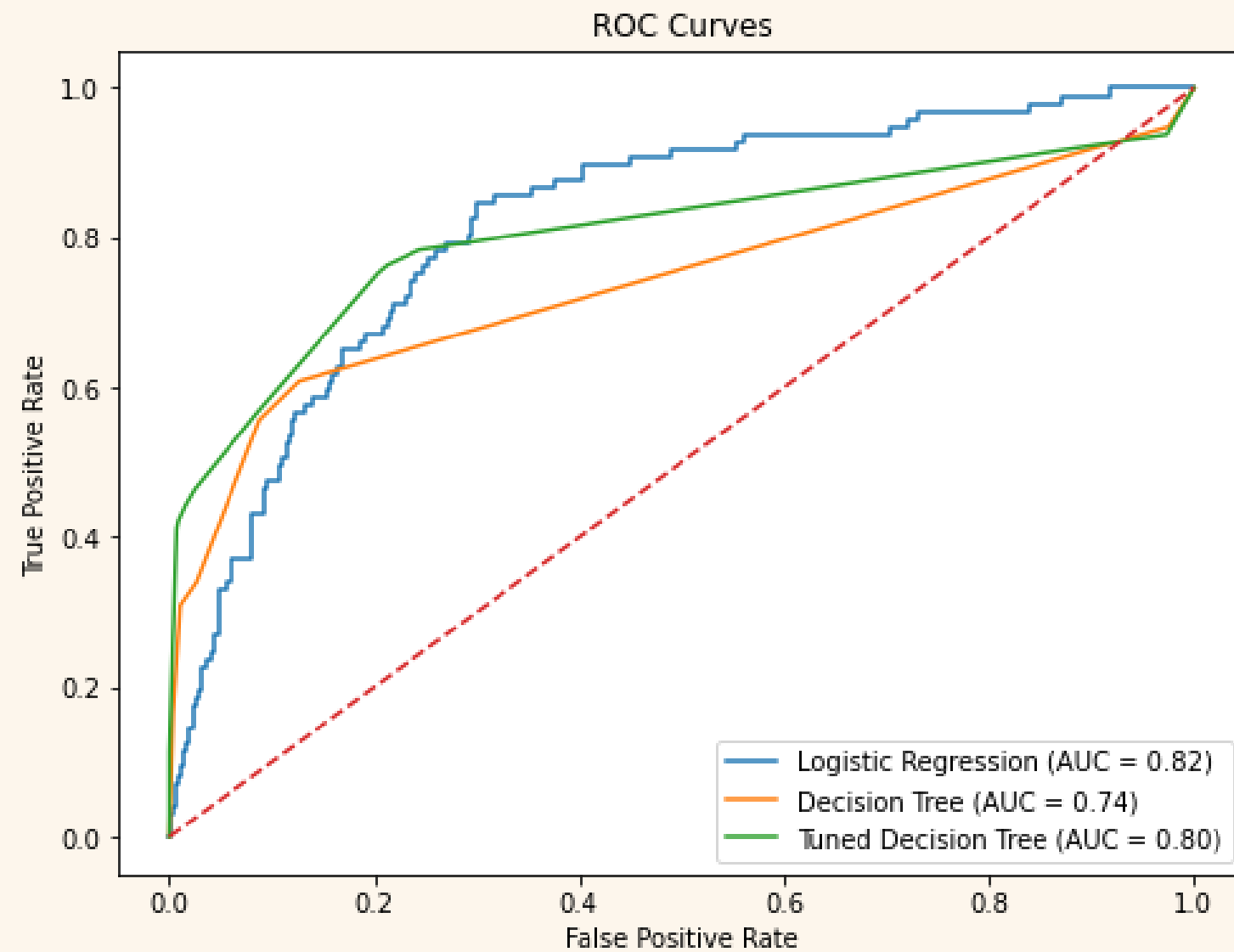
Multiple classification models were evaluated to predict customer churn, including Logistic Regression and Decision Tree approaches, with class imbalance explicitly addressed to avoid biased results. Model selection and tuning prioritized recall to ensure high-risk customers were identified early for proactive retention. All preprocessing and optimization steps were confined to training data to ensure reliable evaluation. Performance was assessed using recall and Precision-Recall AUC, supported by complementary diagnostics, to balance coverage of churn risk with operational efficiency.

Feature importance

Top 10 Feature Importances (Decision Tree)



This chart highlights Total Day Minutes and Customer Service Calls as the primary key indicators that most accurately predict a customer's likelihood of leaving."



With an ROC-AUC of 0.82 and a PR-AUC of 0.60, the model demonstrates strong predictive power and the reliability needed to launch targeted, cost-effective retention campaigns."

The Tuned Decision Tree has the highest recall and PR-AUC

Recommendations

- Deploy the tuned Decision Tree with a recall-focused threshold to identify high-risk churn customers.
- Proactively contact flagged customers with targeted retention offers or plan reviews.
- Focus on customers with high service call rates and distinctive usage patterns.
- Monitor recall, precision, and retention ROI (savings vs. outreach cost).
- Retrain and recalibrate quarterly to address data drift.

conclusion

This project delivers a recall-first churn prediction model suitable for proactive retention. The tuned Decision Tree provides the best recall and PR-AUC among candidates while remaining interpretable for business stakeholders. With , ongoing monitoring, and periodic retraining, the model can be deployed to support timely and cost-effective customer retention strategies.



Thank you

[Github: mburufiona](#)